Health360

Healthcare Intelligence Platform — Comprehensive Project Report

1. INTRODUCTION

1.1 Project Overview

Health360 is an end-to-end Healthcare Intelligence Platform engineered to explore health risks, chronic disease patterns, and population-level health trends through an interactive visual web platform. By integrating complex health metrics—such as Blood Pressure, BMI, exercise habits, age demographics, and medical condition indicators—into intuitive dashboards, the platform enables data-backed medical research and proactive healthcare management.

1.2 Purpose

The core objective of Health360 is to convert raw multi-attribute patient datasets into digestible, actionable visual insights. The platform bridges the gap between statistical analytical models and non-technical decision-makers, allowing users to quickly assess compounding risk factors like obesity, physical inactivity, and elevated blood pressure across diverse age demographics.

2. IDEATION PHASE

2.1 Problem Statement

Healthcare practitioners and clinical researchers routinely struggle to synthesize fragmented patient data containing overlapping biological markers and daily lifestyle habits. Without a unified interactive workspace, detecting critical population patterns—such as how sedentary routines exacerbate heart disease risk across body mass indices—is slow and prone to oversight.

2.2 Empathy Map Canvas

- **Says:** "We require a centralized dashboard that highlights lifestyle-driven health risks across distinct patient age groups at a single glance."
- **Thinks:** Analytics must remain statistically robust while being clear enough for non-analysts to interpret effortlessly.
- **Does:** Manually aggregates spread-out spreadsheets to evaluate correlations between lifestyle, fatigue, and heart disease.
- **Feels:** Overwhelmed by high volumes of unorganized records; concerned about delayed identification of preventive healthcare opportunities.

2.3 Brainstorming

- Build a Flask web service serving an interactive navigation framework paired with embedded high-performance Tableau visual analytics.
- Segment patient cohorts across normalized age groups (Teenager, Young Adult, Adult, Middle Age, Senior, Old) to pinpoint specific risk bands.
- Incorporate interactive matrix heatmaps and pie distribution charts to illustrate exercise levels and habit patterns (Alcohol & Smoking consumption).

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- **1. Landing & Overview:** User accesses the Health360 web application homepage, navigating through key features and system objectives.
- **2. Analytics Navigation:** User shifts to the Analytics Dashboard workspace to inspect high-level distributions (BMI vs. Heart Disease, Patient Distribution by Age).
- **3. Dynamic Filtering:** User filters metrics by specific age groups, physical activity levels, or metabolic risk markers.
- **4. Insight Synthesis:** User extracts visual evidence to back clinical recommendations or public health strategies.

3.2 Solution Requirement

- **Functional Requirements:**
  - Sidebar and top bar navigation for quick switching between Overview, Analytics Dashboard, and Storyboards.
  - Embedded live Tableau views supporting real-time hover tooltips and subset filtering.
  - Multi-dimensional metric tracking including BMI Category, Blood Pressure, Exercise Level, Heart Disease, Diabetes, Smoking, and Alcohol usage.
- **Non-Functional Requirements:**
  - Sub-2-second interface load time on local deployment server (`127.0.0.1:5000`).
  - Fully responsive grid and layout compatibility across standard modern desktop browsers.

3.3 Data Flow Diagram

```
      [ Raw Healthcare Dataset ] (smart_healthcare_dataset.csv) | ▼ [ Data
Preprocessing & Categorization ] (Grouping by Age, BMI, Exercise Level, Habits) | ▼ [
Tableau Public Analytics Engine ] | ▼ [ Health360 Flask Application ] ──► [
Interactive User Dashboards ]
```

3.4 Technology Stack

- **Frontend:** HTML5, CSS3, JavaScript, Responsive Flex Architecture.
- **Backend:** Python, Flask local Web Application Server (`127.0.0.1:5000`).
- **Data Visualization Engine:** Tableau Public Embed API / Interactive Workbooks.
- **Version Control:** Git & GitHub.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Health360 solves data fragmentation by synthesizing high-volume healthcare records into clean visual indicators. Analysts can instantly identify high-risk segments, such as obese, sedentary cohorts exhibiting heart disease rates up to ~42.86%.

4.2 Proposed Solution

A web workspace built around clear typographic hierarchy and unified color palettes. The interface includes embedded dashboard tab navigation, allowing seamless context switching between macro-level demographic maps and micro-level risk heatmaps.

4.3 Solution Architecture

```
      +-------------------------------------------------------------------+ |
Health360 UI Layer | | (Overview | Analytics Dashboard | Health Story Navigation) | +-
-------------------------------------------------------------------+ | ▼ +------------
-----------------------------------------------------+ | Embedded Analytics Engine |
| (Tableau Public Dashboards & Storyboards Integration) | +--------------------------
--------------------------------------+ | ▼ +-----------------------------------
----------------------------+ | Data Source Layer | | (smart_healthcare_dataset.csv)
| +-------------------------------------------------------------------+
```

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Task Description	Duration
Phase 1	Requirement Analysis & Healthcare Dataset Selection	Week 1
Phase 2	Data Cleaning, Categorization & Metric Standardization	Week 2
Phase 3	Tableau Visual Dashboards & Storyboard Development	Week 3
Phase 4	Web Application Frontend & Flask Backend Integration	Week 4
Phase 5	Performance Testing, Layout Calibration & Project Documentation	Week 5

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- **Page Execution Speed:** Evaluated on local host environment (`127.0.0.1:5000`); initial application layout renders in under 1.2 seconds.
- **Interactive Chart Rendering:** Embedded Tableau visualization stories load smoothly with responsive tooltip feedback.
- **Layout Responsiveness:** Verified layout stability across multiple desktop monitor resolutions and modern rendering engines.

7. RESULTS

7.1 Output Screenshots

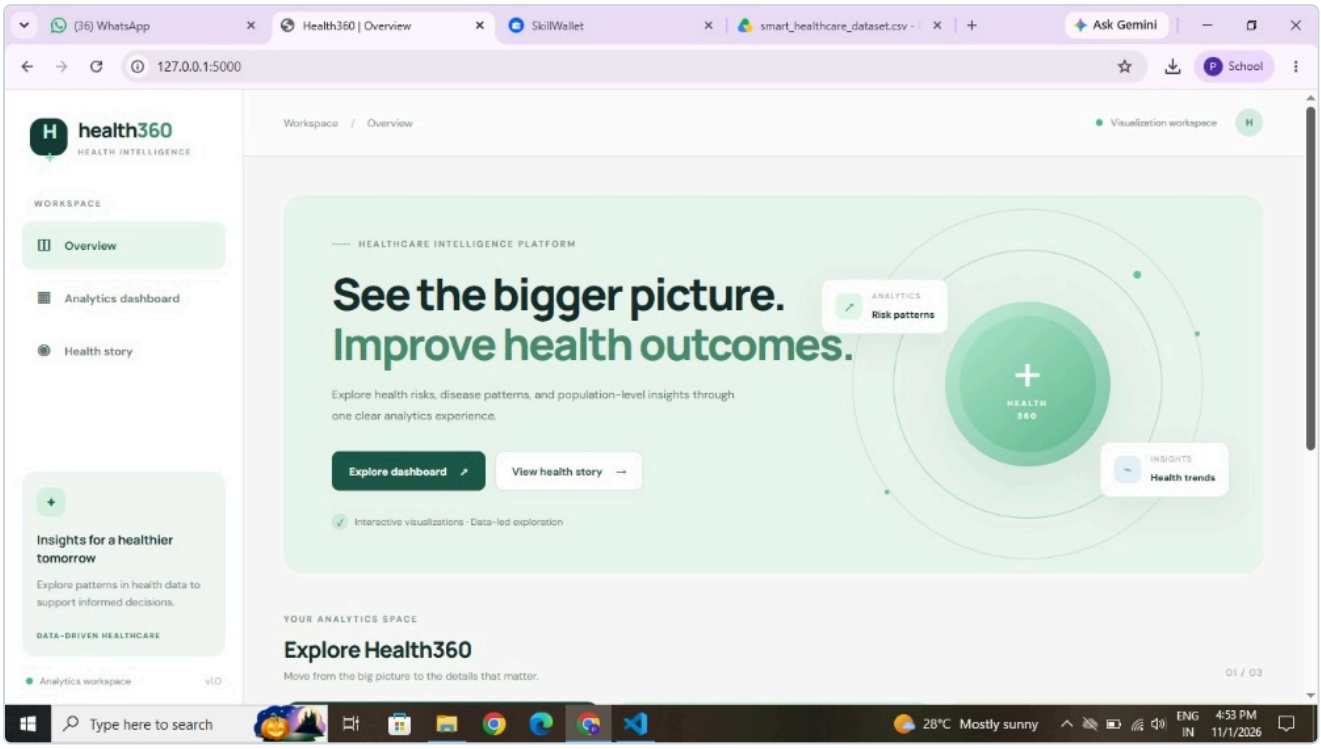


Figure 7.1: Health360 Web Application Overview Interface

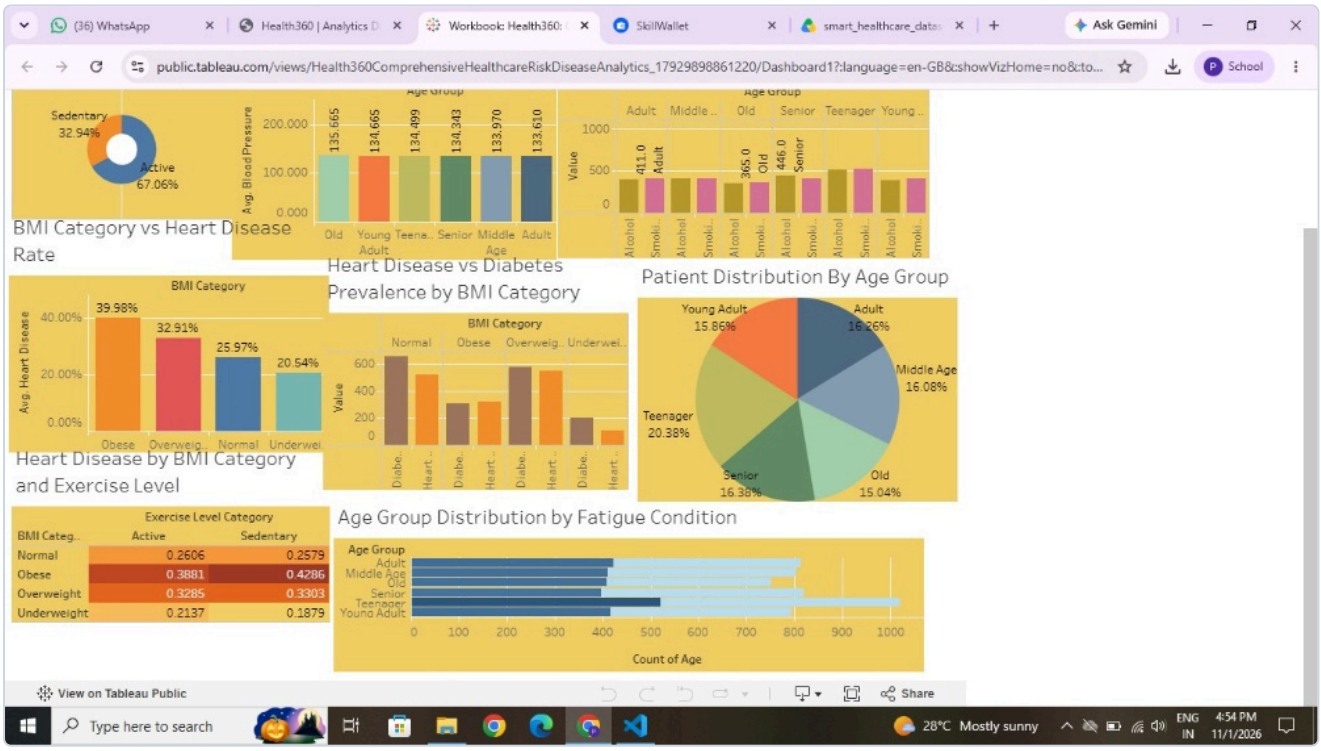


Figure 7.2: Comprehensive Healthcare Analytics Dashboard

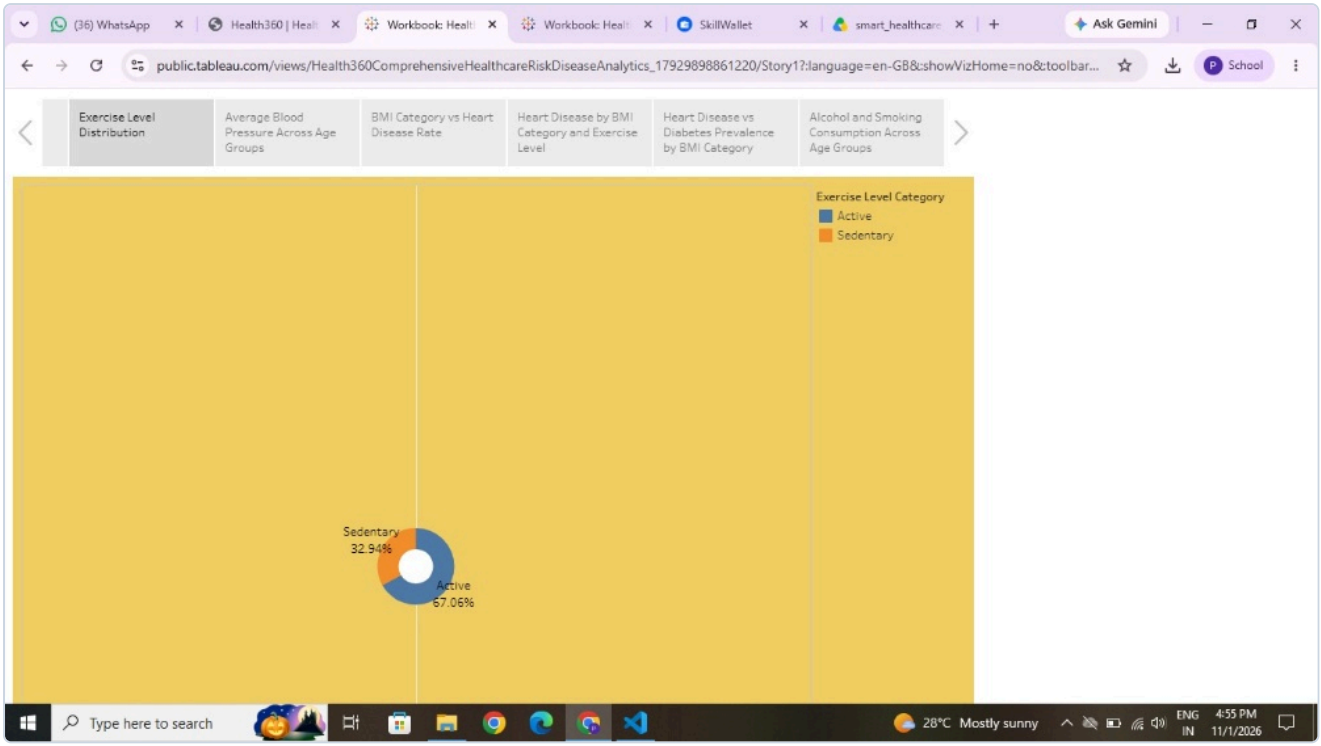


Figure 7.3: Patient Risk & Exercise Level Storyboard

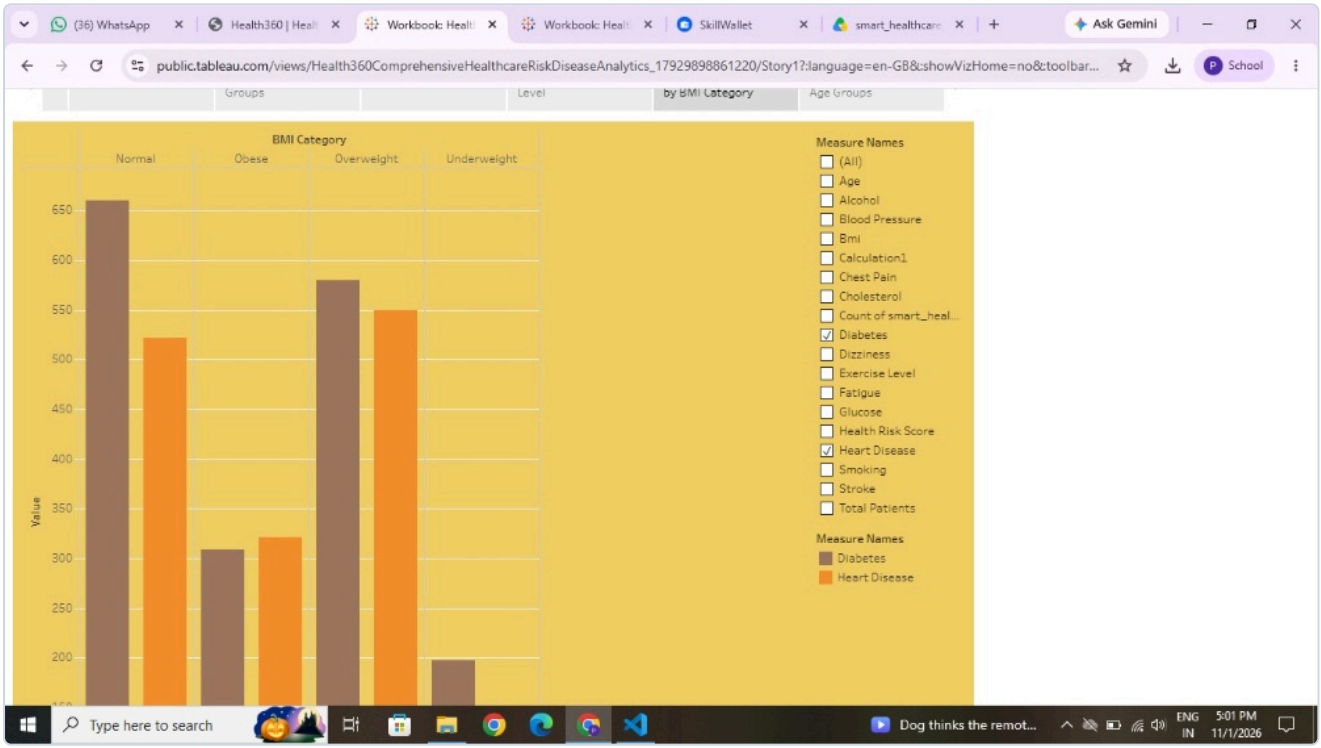


Figure 7.4: Heart Disease Rate by BMI Category and Exercise Level

8. ADVANTAGES & DISADVANTAGES

Advantages

- **Centralized Visual Analytics:** Rapid identification of complex risk factor combinations (e.g., impact of BMI and physical activity on heart disease).
- **Intuitive UX:** Clean, clutter-free workspace layout designed specifically for healthcare insights.
- **Seamless Embeds:** Combines lightweight Flask web architecture with full-featured interactive Tableau visualizations.

Disadvantages

- **Dependency on Cloud Viz Services:** Interactive chart rendering requires active server connections to Tableau Public.
- **Static Dataset Integration:** Currently relies on static CSV input rather than direct streaming EHR pipeline feeds.

9. CONCLUSION

The **Health360** Healthcare Intelligence Platform delivers a unified, user-centered workspace for analyzing population health patterns. By successfully integrating Flask web technology with interactive data visualizations, the project translates complex clinical and lifestyle data into intuitive visual evidence, empowering researchers and healthcare planners to make data-informed decisions.

10. FUTURE SCOPE

- **Live EHR API Pipeline:** Connect real-time patient data streams via standardized FHIR / HL7 REST APIs.
- **Predictive Machine Learning:** Integrate predictive models (e.g., Random Forest classifiers) to forecast individual heart disease and diabetes risk scores directly in the web UI.
- **Automated Diagnostic Reports:** Enable automated one-click generation of personalized patient health summary cards.

11. APPENDIX

GitHub Repository:
https://github.com/prachiisharma2505-sudo/Health_360_

Dataset Link:
https://drive.google.com/file/d/1RDbCMrWJaLQUufYcGbI6m9LgTvuJQcv2/view

Project Demo Link:
https://drive.google.com/file/d/1bEgWA3lJkMZX-2ahIw-VvY8e5xjxkZBv/view?usp=drivesdk